An Explainable Deep Learning Framework for Multi-Class Retinal Disease Classification Using Fundus Images

***Abstract*—Over 400 million individuals globally are affected by different retinal diseases. Although these diseases do not directly cause death, conditions such as age-related macular degeneration and diabetic retinopathy can result in blindness and reduced physical capability, severely affecting life expectancy and quality of life. Early diagnosis is essential, but it is often difficult because it depends on expert eye doctors and advanced diagnostic tools. This study aims to introduce an effective explainable deep learning approach for multi-class retinal disease classification using fundus images from the AMDNet23 dataset. To improve classification performance, three state-of-the-art DL models, EfficientNet-B4, Swin-Tiny, and ConvNeXt-Base, were evaluated individually. Their outputs were further combined using Early Fusion, Late Fusion, and Stacking approaches to identify the most effective combination. In addition, preprocessing techniques, including image resizing, Z-score intensity normalization, bilinear interpolation, and noise reduction, were applied to improve the visual quality of fundus images and enhance model performance. The proposed Swin-Tiny model achieved the best performance among the individual models, with an accuracy of 96.1%. It also outperformed the other individual models in terms of Precision, Recall, and F1-score. The best overall performance was achieved by the Late Fusion approach, with an accuracy of 96.5% and AUC of 99.6%. The proposed explainable model was further analyzed using Grad-CAM to provide visual explanations, revealing that the model focused on clinically relevant areas of the retina. Compared with the reported accuracies of several existing studies, the proposed Late Fusion approach achieved an accuracy improvement of 0.3–5.5%.**

***Index Terms*—Macular Degeneration, Fundus Images, Explainable AI, Ensemble Learning**

I. Introduction

Although retinal diseases are not directly life-threatening, they can indicate other significant health issues. AMD is a primary contributor to significant vision impairment and permanent blindness worldwide, accounting for approximately 8.7% of global blindness [1]. It affects the macula, causing gradual loss of central vision and difficulties in daily activities. Early detection can reduce the negative impacts of numerous diseases through appropriate medical treatment. DL methods can support early detection by developing prediction models that learn patterns from particular datasets [1].

AMD doesn't directly cause death, but it can severely affect the quality of life of older adults. Worldwide, the count of people facing vision-threatening AMD rose from 3.64 million in 1990 to 8.06 million in 2021 and is projected to reach approximately 28.8 million by 2040 [2].

Traditional retinal disease diagnosis relies heavily on ophthalmologists and specialized imaging systems, limiting its use for large-scale screening. Fundus images combined with recent advances in deep learning offer promising potential for automated retinal image analysis, although challenges such as image variability, class imbalance, generalization, and model interpretability remain. Various methods utilizing deep learning for the classification of retinal diseases have been created, including those proposed by A. García-Floriano and Ventura-Molina (2024), A. Abd El-Khalek et al. (2024), and R. Chakraborty and A. Pramanik (2022). While these studies achieved promising predictive performance, most primarily focused on disease classification [3], [4], [6].

The AMDNet23 dataset contains 1,600 color fundus images equally divided into four classes: Normal, Diabetes (Diabetic Retinopathy), Cataract, and AMD, with 400 images per class. The images were collected from six public repositories. This study evaluates EfficientNet-B4, Swin Transformer, and ConvNeXt-Base, along with Early Fusion, Late Fusion, and Stacking ensemble methods to improve model robustness and classification performance.

The main contributions of this study are as follows:

1) Develop a deep learning framework for multi-class retinal disease classification and evaluate CNN-based and Transformer-based architectures.

2) Explore ensemble learning methods, including Early Fusion, Late Fusion, and Stacking, to improve classification accuracy and robustness.
3) Use Explainable AI (XAI) techniques to enhance model transparency and interpretability.

The remainder of this paper is organized as follows: Section II presents the related work, Section III describes the proposed methodology, Section IV presents and discusses the experimental results, and Section V concludes the study.

II. Related Works

In recent years, A. García-Floriano et al. (2024) applied transfer learning with CNNs and achieved 91–94% accuracy on fundus images. However, their study used a small dataset and focused only on binary classification, which limited its generalization to different stages of AMD [3].

R. Chakraborty and A. Pramanik achieved 93–95% accuracy using batch normalization and dropout [4], while N. N. El-Den et al. achieved 96.2% accuracy using a scale-adaptive CNN. However, neither study used Explainable AI (XAI) techniques, limiting the interpretability of their model predictions [5]. A. A. Abd El-Khalek (2024) et al. used handcrafted texture and morphological features with ensemble classifiers. However, this approach required extensive feature engineering and parameter tuning, making the process more complex and limiting its generalization across different datasets [6].

Hybrid, segmentation-based, and OCT-based methods have also been explored for AMD analysis. M. A. Ali et al. proposed the AMDNet23 dataset and a CNN–LSTM model [7]. Li, L. Liang, and Z. Gao (2023) presented an attention-based U-Net for lesion segmentation [8]. A. R. C. Celebi et al. achieved 94–96% accuracy using a CNN with improved preprocessing [9]. Z. Wang et al. used GAN-based data augmentation and improved accuracy by around 4% [10].

Pan et al. [11] proposed a drusen-aware framework combining ResNet50, spatial attention, and LSTM, achieving an AUC above 0.95. However, the framework requires high computational resources. Bhati et al. [12] proposed DKCNet, achieving 96.08% AUC and 94.28% F1-score on the ODIR-5K dataset, which contains an imbalanced image distribution.

Le et al. [13] proposed ViT-AMD, achieving over 95% accuracy, sensitivity, and specificity on the Chula-AMD dataset. XV-AMD [14] used a ViT with SHAP for model explanation and achieved 99.54% accuracy on a custom four-class dataset.

III. Methodology

The goal of this study is to develop a deep learning framework for multi-class retinal disease detection. The proposed method uses a sequential pipeline to achieve high predictive performance and interpretability. Fig. 2 illustrates the proposed framework. First, the dataset is preprocessed, followed by feature extraction and classification using multiple deep learning models. Their outputs are combined to improve prediction robustness. Finally, model performance is evaluated, and Grad-CAM is used to provide visual explanations of the predictions. Each stage is described in the following sections.

A. Dataset Overview

The AMDNet23 dataset is balanced, with 400 images (25%) in each class [15]. Table I provides an overview of the dataset.

TABLE I: Summary of the AMDNet23 Dataset

Description	Value
Data Type	Color Fundus Images
Total Number of Images	1,600
Number of Classes	4 (Normal, Diabetes, Cataract, AMD)
Source Repositories	6 public repositories
Image Variations	Differences in resolution, illumination, contrast, and imaging protocols/cameras

B. Image Preprocessing

To ensure input consistency for deep learning models, retinal images were resized before training and testing. For consistency across all architectures, all fundus images were resized to 256 × 256 pixels before model input. Fig. 1 shows representative fundus image samples from the four diagnostic categories of the AMDNet23 dataset: Normal, Diabetes, Cataract, and AMD [15].



Fig. 1: AMDNet23 fundus image samples from four classes.

Fundus images were first resized to the required input size of each deep learning model using bilinear interpolation:

$$I_r(x', y') = \sum_{x,y} I(x, y)\phi(x' - x)\phi(y' - y) \tag{1}$$

The pixel intensities were then standardized using Z-score normalization:

$$I'(x, y) = \frac{I(x, y) - \mu}{\sigma} \tag{2}$$

Gaussian blur with a 3×3 kernel and randomly sampled $\sigma \in [0.1, 2.0]$ was applied to the training images as a stochastic image-smoothing and noise-reduction augmentation. The resulting preprocessed images were then used for model training, validation, and testing, helping improve image consistency and training stability.

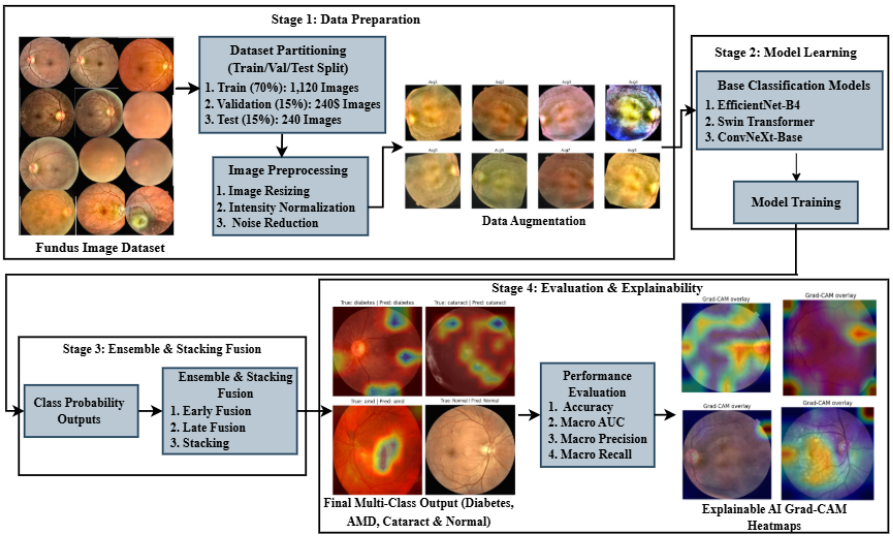


Fig. 2: Overview of the proposed explainable retinal disease classification framework.

C. Data Augmentation

To reduce overfitting caused by limited medical data, data augmentation was applied only to the training set. Augmentation included geometric and photometric transformations while preserving anatomical correctness. Rotation was applied using:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \tag{3}$$

TABLE II: Data Augmentation Parameters used for Training

Augmentation Type	Parameter Range	Applied Set
Rotation	$\pm 15^\circ$	Training
Horizontal Flip	0.5	Training
Vertical Flip	0.5	Training
Random Crop	10%	Training
Brightness	$\pm 20\%$	Training
Contrast	$\pm 20\%$	Training

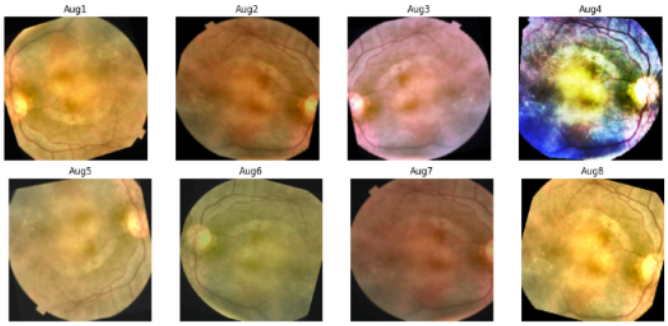


Fig. 3: Augmented images using different transformations.

As shown in Table II, we applied rotation in the range of $\pm 15^\circ$, and performed horizontal and vertical flipping with a probability of 0.5. Random cropping was applied at a rate of 10%, and brightness and contrast were adjusted within a range of $\pm 20\%$. Fig. 3 illustrates data augmentation methods (Aug1–Aug8) used on training fundus images to enhance model generalization.

D. Deep Learning Models

After preprocessing, three DL models, EfficientNet-B4, Swin-Tiny, and ConvNeXt-Base were evaluated individually and in combination using Early Fusion, Late Fusion, and Stacking strategies.

EfficientNet-B4: EfficientNet-B4 uses a compound scaling method that jointly scales the network depth, width, and input resolution. The core scaling principle is defined as:

$$d = \alpha^\phi, \qquad w = \beta^\phi, \qquad r = \gamma^\phi \tag{4}$$

Swin Transformer: It applies window-based multi-head self-attention to capture both local and global contextual relationships. The self-attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \tag{5}$$

ConvNeXt-Base: ConvNeXt-Base is a modern convolutional neural network designed to achieve competitive performance with Vision Transformer-based architectures while retaining the advantages of convolutional networks.

E. Ensemble Fusion Strategies

The complementary information from the three models was combined using three fusion strategies.

Early Fusion: The global pooled features were concatenated to form a 3584-dimensional feature vector, followed by a trainable fully connected layer for 4-class classification:

$$F_{\text{EF}} = F_C \| F_S \| F_E \tag{6}$$

Late Fusion: The 4-class probability outputs of the three models were combined using a weighted sum:

$$P_{\text{LF}} = 0.45P_C + 0.30P_S + 0.25P_E \tag{7}$$

Stacking: The 4-class probability outputs of the three models were concatenated to form a 12-dimensional meta-feature vector:

$$X_{\text{meta}} \in \mathbb{R}^{12} \tag{8}$$

F. Experimental Setup

The proposed models were developed and implemented using PyTorch 1.12.0 and Python 3.9. All deep learning models were trained with an input resolution of 256×256 pixels. Every fold, 70% of the dataset was allocated for training, 15% for validation, and 15% for testing. Early stopping with a patience of 15 epochs was applied based on the validation loss.

G. Explainability Using Grad-CAM

Grad-CAM was employed to improve model interpretability by visualizing the image regions contributing to classification. For class c, the importance weight α_k^c is computed from the gradients of the class score with respect to feature map k:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}, \tag{9}$$

The final Grad-CAM heatmap is generated by applying ReLU to the weighted feature maps.

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right). \tag{10}$$

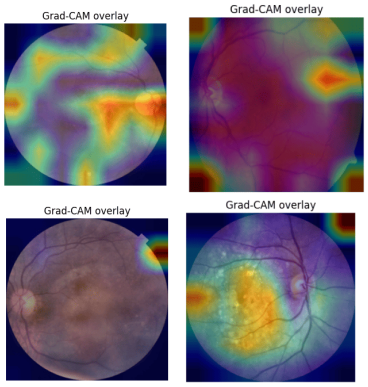


Fig. 4: Grad-CAM heatmaps of retinal regions.

Fig. 4 displays Grad-CAM visualizations showing the model's attention regions for different retinal disease classes.

IV. Results & Discussion

Table III compares Accuracy, Precision, Recall, and F1-Score across the models.

TABLE III: Performance Evaluation of DL Models

Model	Accuracy	AUC	Precision	Recall	F1-score
ConvNeXt-Base	95.0	99.3	96.2	95.6	95.5
EfficientNet-B4	94.9	99.4	95.7	94.3	94.5
Swin-Tiny	**96.1**	**99.5**	**97.1**	**96.8**	**96.5**
Early Fusion	78.0	81.4	80.5	78.4	79.0
Stacking	95.7	99.6	96.5	95.5	96.0
Late Fusion	**96.5**	**99.6**	**96.6**	**95.7**	**96.5**

Based on the results in Table III and Fig. 5, the Late Fusion approach showed the best overall performance, achieving 96.5% accuracy, 99.6% AUC, and strong Precision, Recall, and F1-Score. Among the individual deep learning models, Swin-Tiny achieved the highest accuracy of 96.1%.

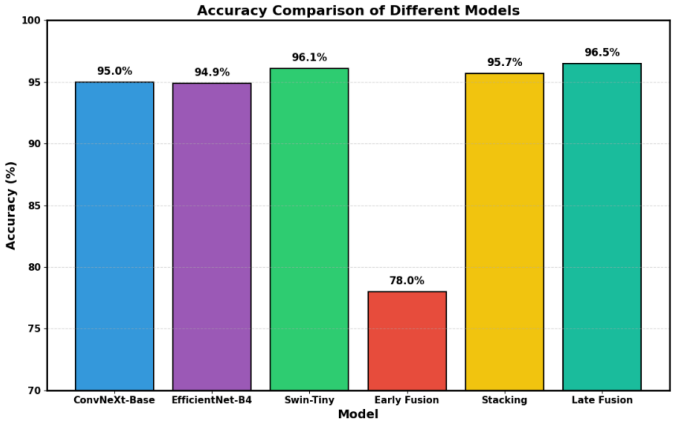


Fig. 5: Acc. comparison of DL models and fusion strategies.

Fig. 6 shows decreasing training and validation loss, while validation accuracy increased from 67.5% to above 85%, indicating consistent learning and good generalization.

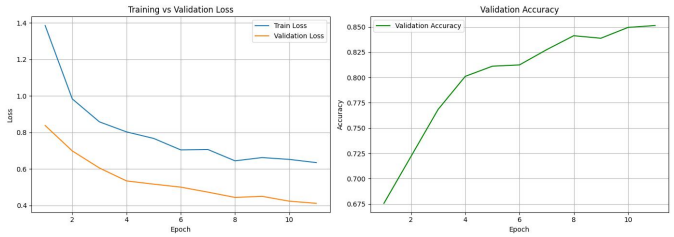


Fig. 6: Training and validation curves of the proposed models.

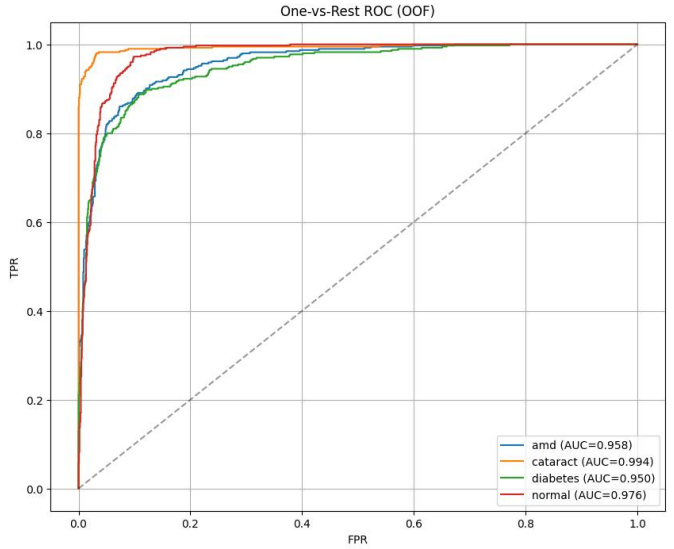
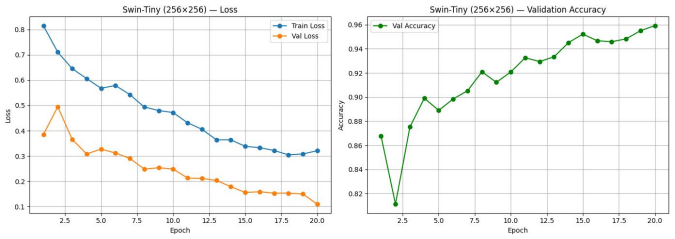
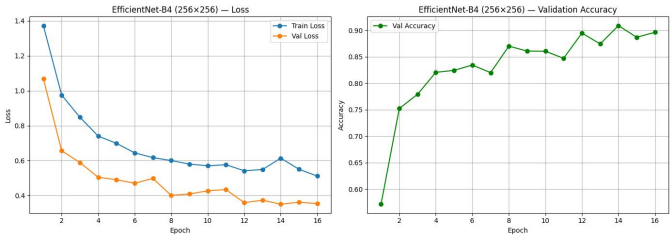


Fig. 7: One-vs-rest ROC curves for retinal disease classes.

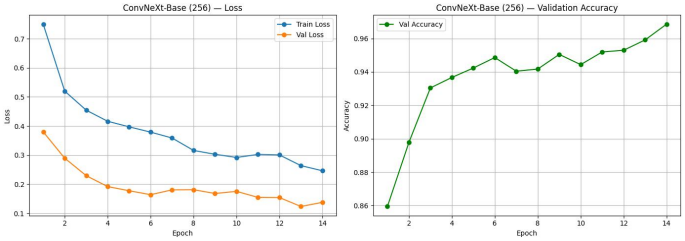
Fig. 7 shows classification performance across all classes. Cataract achieved the highest AUC (0.994), followed by Normal (0.976), AMD (0.958), and Diabetes (0.950). All AUC values exceeded 0.95, indicating strong class discrimination.



(a) Swin Transformer model.



(b) EfficientNet-B4 model



(c) ConvNeXt-Base model

Fig. 8: Training and validation loss, and validation accuracy of the proposed DL models using 256×256 input images.

Fig. 8 learning curves show that all three deep learning models achieved stable convergence during training. ConvNeXt-Base converged rapidly and reached the highest validation accuracy of approximately 97%, while Swin-Tiny achieved over 96% with slight initial fluctuations. EfficientNet-B4 converged quickly but achieved a lower validation accuracy of approximately 90–91%.

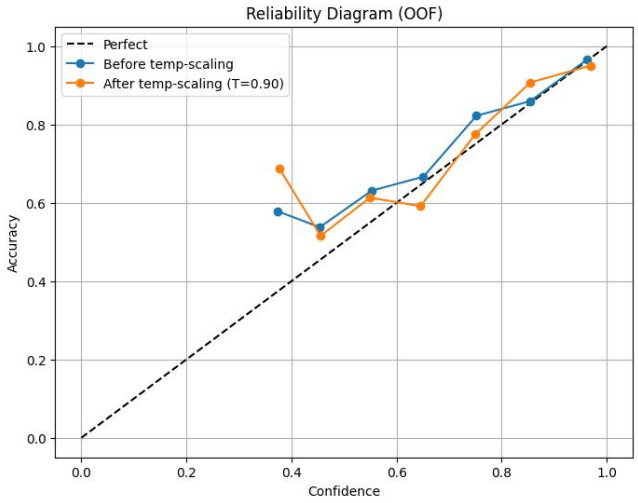


Fig. 9: Reliability plot before and after temperature scaling.

Fig. 9 shows that the model was slightly overconfident before calibration, with an ECE of 0.045. After temperature scaling ($T = 0.90$), the ECE decreased to 0.018 and MCE to 0.052, indicating improved probability calibration.

A. Discussion

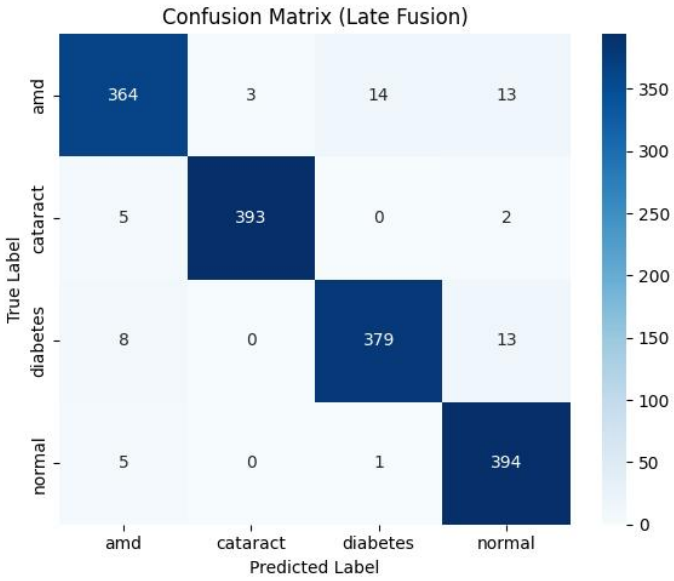
This subsection compares the model with those from earlier studies. Table IV below summarizes the accuracy for the statistical data from this research.

TABLE IV: Comparison of the Proposed and Existing Works

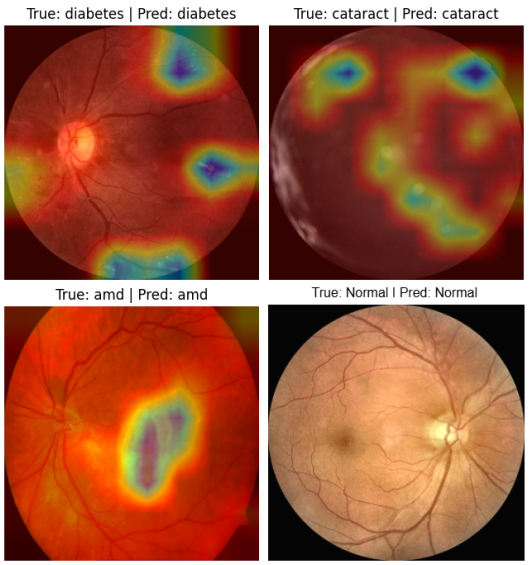
Study/Model	Model Used	Accuracy (%)
A. García-Floriano et al. [3]	CNN with Transfer Learning	91–94
R. Chakraborty et al. [4]	CNN with Batch Normalization and Dropout	93–95
N. N. El-Den et al. [5]	Scale-Adaptive CNN	96.20
A. R. C. Celebi et al. [9]	CNNs	94–96
Le et al. [13]	ViT-AMD	>95
Proposed	ConvNeXt-Base	95.00
Proposed	EfficientNet-B4	94.90
Proposed	**Swin-Tiny**	**96.10**
Proposed	Early Fusion	78.00
Proposed	Stacking	95.70
Proposed	**Late Fusion**	**96.50**

The proposed framework achieved competitive performance compared with several existing studies. A. García-Floriano et al. [3] reported 91–94% accuracy using CNN-based transfer learning, while R. Chakraborty and A. Pramanik [4] achieved 93–95% using batch normalization and dropout. N. N. El-Den et al. [5] reported 96.2% accuracy with a scale-adaptive CNN, and A. R. C. Celebi et al. [9] achieved 94–96% using improved preprocessing. N. T. Le et al. [13] reported over 95% accuracy using ViT-AMD. In comparison, the proposed Late Fusion approach achieved 96.5% accuracy among the evaluated methods, while Swin-Tiny achieved 96.1% among the individual models. The Early Fusion approach achieved 78.0% accuracy, indicating that direct feature concatenation was less effective than probability-based fusion and stacking in this setting.

Fig. 10 (a) displays the confusion matrix for the Late Fusion method. It accurately categorized 393 Cataract, 394 Normal, 379 Diabetes, and 364 AMD images. The confusion matrix indicates strong classification performance across the four retinal disease classes. Fig. 10 (b) presents the Grad-CAM visualizations, highlighting the retinal regions contributing to the classification decisions. The heatmaps show lesion-focused activation for Diabetes, macular activation for AMD, and class-relevant patterns for Cataract and Normal.



(a) Confusion matrix of the late fusion model.



(b) Predictions with Grad-CAM heatmaps.

Fig. 10: Confusion matrix and Grad-CAM-based predictions.

V. Conclusion

In this research, multiple deep learning models were initially assessed for classifying various retinal diseases. Different experimental methods, such as image resizing, Z-score standardization, data augmentation, and fusion techniques, were subsequently utilized to evaluate the effectiveness of the suggested framework. Among the individual models, Swin-Tiny achieved an accuracy of 96.1%, whereas Late Fusion achieved the best overall performance, with 96.5% accuracy. Furthermore, Grad-CAM visualizations showed that the model focused on clinically relevant retinal regions, such as blood vessels, drusen areas, and the macula, enhancing the transparency and trustworthiness of its predictions. Future work will validate the framework on larger, multi-center datasets and explore multimodal learning using fundus images with OCT or clinical data. Overall, combining multi-model learning, fusion strategies, and explainable AI provides accurate and interpretable retinal disease classification, supporting future AI-based retinal screening systems. In the future, we will launch the web application on a cloud platform that is openly available and can be assessed by actual users.

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